

CSE302 | Database Systems (Reference)

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Key Constraints

- Keys are entity sets used to identify entities within their respective sets uniquely.
- Key constraints play a vital role in ensuring the uniqueness of data within a table.

Key

- A key in DBMS is an attribute or a set of attributes that help to uniquely identify a tuple (or row) in a relation (or table).
- There are broadly seven types of keys in DBMS.
 1. Primary Key
 2. Candidate Key
 3. Super Key
 4. Foreign Key
 5. Composite Key
 6. Alternate Key
 7. Unique Key

An example

- Consider a table named 'Student'

Student

Name	Address	Age
Raaj	Dhaka	21
Nuhash	Chittagong	23
Raaj	Dhaka	21

- How can we uniquely identify two rows in this table?

An example

- We can add a column named roll

Student

Roll	Name	Address	Age
1	Raaj	Dhaka	21
2	Nuhash	Chittagong	23
3	Raaj	Dhaka	21

- What are the other columns we can use instead of 'Roll'?

Registration_no, NID_no, Phone_no and so on...

Observation

- {Roll, Registration_no, NID_no, Phone_no, email ... }
- We call the collection of all the attributes that can uniquely identify any two tuples or rows in a table, **Candidate key**
- There can be more than one candidate keys in a table.
- **The Primary Key of a table is selected (the most appropriate one) from one of the candidate keys.**
- The rest of the keys that remain after choosing the primary key are known as **alternate key**

Primary Key

- The primary key constraint is used to uniquely identify each record (row) in a database table.
- It enforces the uniqueness of values in the specified column or set of columns and ensures that no two records have the same primary key value.

```
CREATE TABLE Persons (  
    ID int NOT NULL,  
    LastName varchar(255) NOT NULL,  
    FirstName varchar(255),  
    Age int,  
    PRIMARY KEY (ID)  
);
```

```
CREATE TABLE Persons (  
    ID int NOT NULL PRIMARY KEY,  
    LastName varchar(255) NOT NULL,  
    FirstName varchar(255),  
    Age int  
);
```

- Here ID in the Persons table is defined as primary key

Unique Key

- The **UNIQUE** constraint ensures that all values in a column are different.
- The unique constraint ensures that values in a specified column or columns are unique across all rows in a table, but it doesn't require them to be the primary key.
- Both the **UNIQUE** and **PRIMARY KEY** constraints provide a guarantee for uniqueness for a column or set of columns.
- A **PRIMARY KEY** constraint automatically has a **UNIQUE** constraint.
- However, you can have many **UNIQUE** constraints per table, but only one **PRIMARY KEY** constraint per table.

Unique Key

```
CREATE TABLE Persons (  
    ID int NOT NULL UNIQUE,  
    LastName varchar(255) NOT NULL,  
    FirstName varchar(255),  
    Age int  
);
```

```
CREATE TABLE Persons (  
    ID int NOT NULL,  
    LastName varchar(255) NOT NULL,  
    FirstName varchar(255),  
    Age int,  
    UNIQUE (ID)  
);
```

- Here The LastName column is defined as it will contain all unique values but it is not necessarily the primary key

Super Key

- Combination of all possible attributes that includes at least one candidate key together which can uniquely identify two tuples in a table is super key

Student

Roll	Name	email	Age
1	Raaj	abc@g.com	21
2	Nuhash	def@g.com	23
3	Raaj	gdr@g.com	21

Super Key

- Combination of all possible attributes that includes at least one candidate key together which can uniquely identify two tuples in a table is super key
- Super Key is the superset of a candidate key.
- Super key = Candidate key + 1 or more attribute

Student

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Student

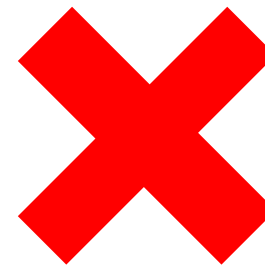
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Primary Key vs Unique Key

Feature	Primary Key	Unique Key
Purpose	Main row identifier	Extra unique constraint
NULL allowed	✗ No	✓ Yes
Auto increment	✓ Yes (usually)	✗ No
Count per table	Only one	Multiple
Can change value	Technically yes, not recommended	Yes

Super Key vs Candidate Key

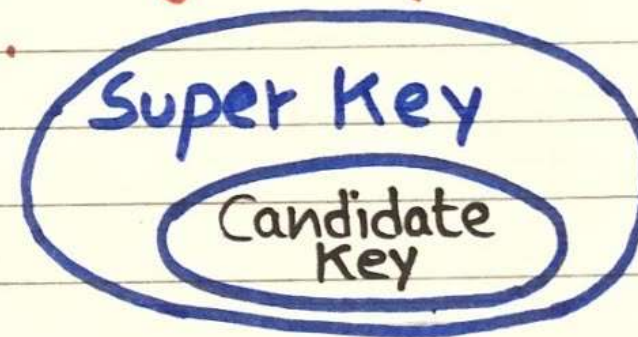
Difference Between Super Key and Candidate Key

Super Key

A super key is a set of collection of one or more than one attributes that can identify data uniquely.

Candidate Key

Candidate Key is a subset of a super Key.



Foreign Key

- a reference link between tables that are known as the primary (or parent) and foreign (or child) tables. The foreign table references one or more columns (the primary key, which can be one or more columns) in the primary table; that is how the link is created.
- **A FOREIGN KEY is a field (or collection of fields) in one table, that refers to the PRIMARY KEY in another table.**

Foreign Key

- a FOREIGN KEY on the "user_id" column when the "Orders" table is created
- FOREIGN KEY (columnName) REFERENCES tablename(columnName)

users			orders		
user_id	email	name	order_no	user_id	product_sku
10	sadio@example.com	Sadio	93	11	123
11	mo@example.com	Mohamed	94	11	789
12	rinsola@example.com	Rinsola	95	13	789
13	amalie@example.com	Amalie	96	10	101

Foreign Key

users		
user_id	email	name
10	sadio@example.com	Sadio
11	mo@example.com	Mohamed
12	rinsola@example.com	Rinsola
13	amalie@example.com	Amalie

orders		
order_no	user_id	product_sku
93	11	123
94	11	789
95	13	789
96	10	101

books		
product_sku	title	price
123	Aurora	15
456	Blind Lake	10
789	Invisible Planets	25
101	The Sparrow	15

97	NULL	101
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Advantage of Foreign Key

- Self study

Primary key	Foreign key
It must contain unique values.	It can contain duplicate values.
It cannot contain null values.	It can contain null values.
A database can have only one primary key.	A database can have more than one foreign key.
It is used to identify the records in a table uniquely.	It is used to make a relation between two tables.

Thank You